



GAMES OF LEARNING

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Slogan:	<i>Gaming minds. Bright Futures.</i>

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Disaster Recovery Plan (DRP) – Games for Learning (GfL)

A Disaster Recovery Plan (DRP) for a gaming company serves as a strategic blueprint to minimize downtime, data loss, and revenue impact during and after unexpected disruptions (Snedaker, 2013).

1. Purpose

The purpose of this Disaster Recovery Plan (DRP) is to provide a systematic and orderly manner of recovering Games for Learning's (GfL) mission-critical IT systems in the event of a disruption (Snedaker, 2013). The file server and database server located at the main office are essential to the daily operations of the company. The file server houses centralized document management, and the database server hosts the customer catalog and client base that are central to business continuity (Snedaker, 2013).

This plan ensures that everyone understands their part in the event of a disaster, backups are reliable, and recovery procedures can get operations up and running with minimal downtime and data loss. GfL would face disastrous operational, financial, reputational, and legal losses without an adequate DRP (Snedaker, 2013).

2. Objectives

The primary objectives of this DRP are:

- Minimize Downtime: Reduce the period servers and business-critical systems are unavailable in case of a disaster (Snedaker, 2013).
- Minimize Data Loss: Ensure backups and recovery processes protect business-critical data, with no greater than 24 hours of data loss (Recovery Point Objective).
- Define Roles and Responsibilities: Determine what IT personnel, managers, and vendors are responsible for implementing each step of the recovery process.
- Standardize Procedures: Develop detailed, step-by-step procedures to be followed during recovery (Snedaker, 2013).
- Ensure Business Continuity: Maintain critical business functions such as access to client data, employee records, and shared resources despite adverse situations.
- Compliance and Governance: Ensure adherence to industry best practices, IT governance, and data protection laws such as South Africa's POPIA (Protection of Personal Information Act) (Snedaker, 2013).

The overall objective of GFL's Disaster Recovery Plan is to enable the company to reinstate critical operations and restore critical data and systems in the least amount of time practicable following a disruption, with minimum operational, financial, and reputational exposure.

The objective serves to help the company achieve its purpose of furnishing uninterrupted, quality gaming and maintaining long-term user trust and satisfaction, even in the event of unexpected technical or environmental disruptions.

3. Scope

This design is for GfL's primary office site IT infrastructure and includes:

- File Server: Provides storage for internal documentation, home directories, and shared files (Whitman and Mattord, 2017).

- Database Server: Contains catalog and client details, supporting client management and e-learning systems.
- Supporting Systems: Network hardware, firewall configurations, and authentication services that provide access to the servers (Whitman and Mattord, 2017).

Exclusions:

- End-user laptops and desktops (only covered if directly impacted by a server outage) (Whitman and Mattord, 2017).
- The new building's network infrastructure.
- Third-party services such as Internet lines from Telkom (though escalation is in place) (Whitman and Mattord, 2017).

The scope of a Disaster Recovery Plan determines the boundaries within which the plan operates. It specifies what systems, departments, processes, and types of disasters are covered, and ensures that all parties understand what the plan is covering and not.

For GfL, a web gaming company operating online platforms and services, the DRP scope is designed to encapsulate all business-critical processes and online infrastructure supporting product delivery, user experience, and business continuity.

4. Risk Identification

Risk identification enables GfL to be aware of the potential threats that can lead to disruption of IT services. The risks are categorized as natural, technical, human, and external risks.

4.1 Risk Categories

Risk Type	Example	Potential Impact	Mitigation Strategy
Natural Disasters	Flood, fire, lightning, storms	Destruction of IT infrastructure, prolonged downtime	Off-site backups, cloud storage, insurance, fire suppression

			systems (Wallace and Webber, 2017).
Power-Related Risks	Load shedding, UPS failure, generator malfunction	Server downtime, data corruption	UPS systems, backup generators, energy monitoring
Hardware Failure	Disk crashes, RAID controller failure	Loss of access to servers, partial or total data loss	RAID configurations, spare parts, warranties, hardware monitoring (Wallace and Webber, 2017).
Cybersecurity Risks	Ransomware, phishing, unauthorized access	Data breaches, corruption, reputational damage	Firewalls, antivirus, intrusion detection, staff training
Network Failures	ISP outage, router/firewall malfunction	No access to internet, intranet unavailable	Dual ISP connections, failover routers, vendor support (Wallace and Webber, 2017).
Human Error	Accidental deletion, misconfigurations	Data loss, downtime, mismanagement	Regular training, access control, documented SOPs

4.2 Risk Analysis Matrix

Likelihood	Impact	Risk Level
High	High	Critical – Requires immediate controls (cyberattack, power outage) (Wallace and Webber, 2017).
Medium	High	Major – Needs proactive monitoring (hardware failures)
Low	High	Moderate – Low chance but severe (natural disasters)
Medium	Medium	Moderate – Manageable (human error) (Wallace and Webber, 2017).
High	Low	Minor – Acceptable with basic controls (printer issues) (Wallace and Webber, 2017).

- Risk identification is the critical first step in creating an effective Disaster Recovery Plan. By proactively envisioning a wide variety of threats ranging from technological failures to environmental disasters, Edunova's recovery strategies based on real scenarios and recovery plans are comprehensive and actionable (Wallace and Webber, 2017).

5. Backup Strategy

- The backup strategy safeguards vital information from all the risks identified (Wallace and Webber, 2017).

5.1 Backup Frequency

- Daily Backups: Full backup of file and database servers once a day after business hours (11 PM).
- Weekly Backups: Weekly consolidated full backup offsite (Wallace and Webber, 2017).
- Monthly Backups: Archived to long-term encrypted storage, kept for 6 months.

5.2 Storage Locations

- Primary On-Site Backup: Network Attached Storage (NAS) device configured with RAID-6 redundancy (Wallace and Webber, 2017).
- Secondary Off-Site Backup: Encrypted cloud storage (Azure/AWS S3).
- Removable Media: External encrypted hard drives for compliance archiving, stored in a fireproof safe (Wallace and Webber, 2017).

5.3 Retention Policy

- Daily backups retained for 30 days.
- Weekly backups retained for 12 weeks.
- Monthly backups retained for 6 months.

5.4 Testing & Validation

- Quarterly test restores performed to verify data integrity.
- Backup logs checked daily by IT staff.

- Any failed jobs rerun immediately (Wallace and Webber, 2017).

6. Recovery Procedures

Recovery processes are divided into preparation, recovery, validation, and return-to-service steps (Snedaker, 2013).

6.1 Preparation

- Disaster discovery: IT staff recognize disaster (e.g., ransomware attack, hardware failure).
- Invoke DRP: IT Manager activates disaster recovery mode.
- Alert stakeholders via communication channels (Snedaker, 2013).

6.2 Recovery of File Server

- Replace faulty hardware or virtualize recovery environment (Snedaker, 2013).
- Restore full back up from NAS (if available).
- Restore from cloud storage if NAS is unavailable.
- Confirm file permissions and shares.
- Reconnect employees to home folders and shared drives (Snedaker, 2013).

6.3 Database Server Recovery

- Provide a new database server (physical or VM) (Snedaker, 2013).
- Restore SQL database from most current full backup.
- Apply incremental logs if any.
- Test application connections.
- Verify catalog and customer records (Snedaker, 2013).

6.4 Validation

- Internal testing is performed by IT staff.
- Data integrity and access are confirmed by selected end-users.
- Restoration results are documented (Snedaker, 2013).

6.5 Return-to-Service

- Resume normal business activities.
- Monitor systems for 24 hours.
- Create incident report (Snedaker, 2013).

7. Communication Plan During Disaster

7.1 Roles & Responsibilities

- IT Manager (Incident Commander): Declares disaster, directs recovery, liaises with management (Wallace and Webber, 2017).
- Network Administrator: Executes recovery tasks, restores servers, tests connectivity.
- Helpdesk Staff: Liaises with end-users, monitors issues, updates service desk.
- Vendors: Provide hardware, ISP, and software support as required (Wallace and Webber, 2017).

7.2 Communication Channels

- Primary: Internal email system.
- Secondary: SMS/phone trees if no email.
- Tertiary: Company WhatsApp group or intranet page (Wallace and Webber, 2017).

7.3 Stakeholder Updates

- Executives: Hourly update during crisis.
- Employees: Daily update until full services resumed.
- Customers: Public update only if customer-facing systems affected (Wallace and Webber, 2017).

8. Review & Maintenance

8.1 Review Schedule

- DRP reviewed every 6 months.
- Updates finished after any significant disaster, infrastructure change, or IT audit (Snedaker, 2013).

8.2 Training & Awareness

- IT staff trained in implementing DRP.
- Disaster simulation exercises carried out every year to ensure response readiness.
- Awareness training for personnel to reduce human error risks (Snedaker, 2013)

8.3 Continual Improvement

- Lessons learned documented following every incident.
- Metrics tracked: downtime, recovery success rate, backup success rate (Snedaker, 2013).

8.4 Documentation

- Version-controlled DRP stored safely and retrievable for key staff
- Alterations recorded with justification for change
- Training provided to all members of the team on a yearly basis (Snedaker, 2013).

9. Approval

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